

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO4: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Mock Interview

Duration: 1hr

Code of Conduct:

I ALLAPAREDDY JAHNAVI(192472214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Jahnavi

1. A machine learning engineer applies Reinforcement Learning to solve sequential decision-making problems. Explain how an agent learns optimal behavior through interaction with the environment.

Reinforcement Learning is a learning approach in which an **agent learns by interacting with an environment** rather than being given the correct actions directly.

The learning process works as follows:

1. The agent observes the current state of the environment.
2. Based on its policy, it selects an action.
3. The environment responds with a reward and a new state.
4. The agent evaluates whether the action was beneficial.
5. It updates its policy or value estimates based on the received experience.
6. This process is repeated to maximize the cumulative future reward.

The main components are state, action, reward, policy, and environment.

For example, in a game-playing AI, the state represents the current game situation, the action is a possible move, and the reward may be positive for winning and negative for losing. Through repeated trials, the agent gradually learns which actions produce better long-term results.

Therefore, the agent learns optimal behavior through trial and error, reward feedback, exploration, and policy improvement.

2. If you are asked to design a Reinforcement Learning solution for a real-time system such as ride-sharing or delivery optimization, explain how you would formulate the problem using a Markov Decision Process.

I would formulate the ride-sharing or delivery problem as a **Markov Decision Process (MDP)** consisting of **states, actions, transition probabilities, rewards, and a policy**.

For a ride-sharing system:

- **State:** Current vehicle location, passenger demand, traffic condition, vehicle availability, and time.

- **Action:** Assign a vehicle to a passenger, move to a particular location, or accept/reject a request.
- **Reward:** Positive reward for completing rides efficiently and negative reward for delays, fuel consumption, or cancellations.
- **Transition:** The environment changes according to traffic, new ride requests, completed trips, and vehicle movements.
- **Policy:** Determines the best action for each state.

The objective is to maximize the expected cumulative reward:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

where γ is the discount factor.

For example, if demand is high in one area, the agent can learn to position available vehicles there in advance. Thus, the MDP provides a mathematical framework for making sequential decisions under changing conditions.

3. Compare model-free and model-based Reinforcement Learning approaches. Explain their working principles and how they handle uncertainty.

The main difference is whether the agent learns or uses an explicit model of the environment.

Feature	Model-Free RL	Model-Based RL
Environment model	Not required	Learns or uses a model
Learning	Directly learns policy/value	Learns dynamics and plans
Planning	Limited	Possible through simulation
Sample efficiency	Usually lower	Usually higher
Computational requirement	Generally lower	Higher due to planning
Model error	Not applicable	Can affect performance

Model-Free RL directly learns a policy or value function from experience. Examples include Q-learning, SARSA, and policy-gradient methods.

Model-Based RL learns an approximate model of how actions affect future states and rewards. It can then simulate possible future outcomes before taking an action.

Regarding uncertainty, model-free methods handle uncertainty mainly through exploration and repeated experience, while model-based methods can explicitly represent or estimate uncertainty in the learned dynamics.

For example, a robot can use model-based RL to predict the result of different movements before actually executing them, reducing unnecessary real-world trials.

4. A recommendation system using Reinforcement Learning is underperforming. Explain how reward sparsity and hyperparameter tuning can be diagnosed and improved.

I would first analyze whether the problem is caused **by** insufficient reward feedback, poor exploration, or unsuitable hyperparameters.

Reward sparsity occurs when the agent receives meaningful rewards only occasionally. For example, a recommendation system may receive a reward only when a user purchases an item.

To improve this:

- Use suitable intermediate rewards such as clicks, time spent, or engagement.
- Design the reward carefully so that it represents the actual objective.
- Encourage exploration so that the agent discovers useful recommendations.
- Monitor reward distribution and the percentage of episodes receiving non-zero rewards.

I would also tune important hyperparameters such as:

- Learning rate
- Discount factor
- Exploration rate
- Batch size
- Network parameters, if a neural policy is used

I would perform controlled experiments and **compare** average reward, convergence speed, and recommendation performance.

For example, if the learning rate is too high, training may become unstable. If exploration is too low, the system may repeatedly recommend the same items and fail to discover better choices.

5. If you are implementing an RL-based game-playing AI, describe how you would design the learning pipeline and ensure training stability.

I would design the pipeline in the following stages:

**Environment → State Observation → Policy → Action → Reward → Experience
Storage → Learning → Policy Update → Evaluation**

First, I would define the game state and available actions. The agent selects actions using its policy and receives rewards based on game outcomes.

A typical training pipeline would include:

1. Initialize the policy and learning parameters.
2. Let the agent interact with the game environment.
3. Store experiences such as state, action, reward, and next state.
4. Train the policy using collected experiences.
5. Periodically evaluate the agent without exploration.
6. Continue training until performance stabilizes.

For training stability, I would use an appropriate learning rate, controlled exploration, reward normalization when necessary, mini-batch training, and experience replay for suitable algorithms.

I would also monitor average reward, win rate, loss, and convergence. If the performance becomes unstable, I would reduce the learning rate or adjust the exploration strategy.

6. A company wants to use Reinforcement Learning for dynamic pricing. Explain how the agent learns from interaction and improves pricing decisions over time.

In dynamic pricing, the RL agent learns which price produces the best long-term business outcome.

The problem can be defined as:

- **State:** Current demand, inventory, time, customer behavior, competitor prices, and market conditions.

- **Action:** Select a price from a predefined range.
- **Reward:** Profit, revenue, sales volume, or a combination of business objectives.
- **Policy:** Determines the price to select for each market condition.

For example, if demand increases during peak hours, the agent may learn that a higher price can increase revenue while still maintaining acceptable demand.

The agent initially explores different pricing strategies. After observing the resulting sales and rewards, it updates its policy. Over many interactions, it learns which prices are most effective under different conditions.

A suitable reward function should also include business constraints so that the agent does not maximize short-term revenue at the expense of customer satisfaction or long-term profitability.

7. If an RL model is not converging properly, explain how you would identify and fix issues related to learning rate and exploration strategy.

If an RL model is not converging, I would first examine the reward curve, loss values, policy behavior, learning rate, and exploration rate.

A learning rate that is too high can cause large parameter updates, resulting in unstable training. I would reduce the learning rate or use a suitable learning-rate schedule.

If the learning rate is too low, learning may become extremely slow. In that case, I would gradually increase it.

I would also analyze the exploration strategy:

- **Too much exploration:** The agent continues taking random actions and does not exploit learned knowledge.
- **Too little exploration:** The agent may get stuck with a poor strategy and fail to discover better actions.

For an ϵ -greedy strategy, I would start with a relatively higher ϵ and gradually reduce it:

$$\epsilon_{t+1} < \epsilon_t$$

This allows the agent to explore initially and exploit better actions later.

Finally, I would compare different settings experimentally and select the configuration that provides stable reward improvement and consistent convergence.

8. A robotics system uses Reinforcement Learning for navigation. Explain how the environment and agent interaction leads to optimal path learning.

In robot navigation, the robot acts as the RL agent, while the physical or simulated surroundings form the environment.

The state may contain:

- Robot position
- Distance from obstacles
- Sensor readings
- Destination position
- Current direction or velocity

The actions may include:

- Move forward
- Turn left
- Turn right
- Stop

The reward function guides the robot toward the destination. For example:

- Positive reward for moving closer to the target.
- Large positive reward for reaching the target.
- Negative reward for collisions.
- Small negative reward for unnecessary movement or excessive time.

The robot repeatedly interacts with the environment and learns which actions produce higher cumulative rewards.

Over time, the policy learns a sequence of actions that avoids obstacles and reaches the destination efficiently. Therefore, reward feedback combined with repeated interaction enables the robot to learn an optimal or near-optimal navigation policy.

9. If you are asked to improve an RL model with delayed rewards, explain techniques to handle credit assignment problems.

The credit assignment problem occurs when an important action happens much earlier than the reward that indicates its success. The agent must determine which previous actions contributed to the final outcome.

For example, in a game, the agent may receive a winning reward only after many actions. It is difficult to determine which individual actions were responsible for the win.

I would use the following techniques:

1. **Discounted returns** – assign appropriate importance to future rewards.
2. **Temporal-Difference learning** – update value estimates using subsequent states.
3. **Eligibility traces** – distribute credit across recently visited states and actions.
4. **Reward shaping** – provide meaningful intermediate rewards when appropriate.
5. **Advantage estimation** – determine whether an action performed better than the expected value.
6. **Multi-step returns** – use several future rewards instead of only one-step information.

For example, in robot navigation, instead of giving reward only when the robot reaches the destination, intermediate rewards for moving closer to the target can provide better learning signals.

These techniques help the agent identify which actions contributed to long-term success.

10. A financial firm uses Reinforcement Learning for trading decisions. Explain how uncertainty and risk can be handled in the learning process.

Financial markets are highly uncertain because prices are affected by market trends, news, economic conditions, and unpredictable events. Therefore, an RL trading system should consider both expected return and risk.

The state can include:

- Historical prices
- Trading indicators

- Portfolio value
- Current holdings
- Market conditions

The actions can be buy, sell, or hold.

The reward should not simply maximize profit. It can also penalize excessive risk, transaction costs, and large losses.

For example:

$$\mathbf{Reward}_t = \mathbf{Profit}_t - \lambda \cdot \mathbf{Risk}_t - \mathbf{Transaction Cost}_t$$

where λ controls the importance of risk.

To handle uncertainty, I would use sufficient exploration during training, evaluate the policy on different market periods, and test it using unseen data. Risk measures such as maximum drawdown, volatility, and risk-adjusted return can also be monitored.

Most importantly, the system should include risk limits and not be allowed to take unlimited positions. Thus, RL can optimize trading decisions while explicitly considering uncertainty, risk, and long-term performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately